Informative Stock Dependency Learning via Causal-Invariance-Aware Spatio-Temporal Perturbations

Supplementary Material

I. Related Work

A. Time Series Forecasting and Stock Prediction

Time series forecasting aims to capture temporal dependencies from historical observations for future inference [1]. Early studies mainly rely on statistical models, while recent research has increasingly adopted deep neural networks, including RNNs [2], Transformers [3], [4], and MLP-based architectures [5], [6], to model complex nonlinear temporal patterns. These models have also been widely applied to stock prediction, where historical price indicators are used to infer future prices, returns, or movement trends. To further improve prediction, recent studies incorporate complementary information such as market indices, sector information, macroeconomic variables, and financial news [7], [8], [9], [10]. Despite their effectiveness, these approaches mainly focus on stock-specific temporal dynamics and have limited ability to explore interactions among related stocks.

B. Inter-Stock Dependency Modeling

Since stocks are interconnected through various economic and market interactions, increasing efforts have focused on modeling inter-stock dependencies for prediction [11]. Early approaches construct static stock graphs using predefined financial knowledge, such as industry affiliations and shareholding relationships, and employ graph neural networks to aggregate information from related stocks [12], [13]. To capture evolving market interactions, subsequent studies explore implicit and dynamic inter-stock relations through correlation measures, attention mechanisms, and adaptively learned graph structures [14], [15], [16], [17]. Despite these advances, existing methods primarily focus on modeling the strength of inter-stock dependencies, while weakly related or irrelevant stocks may still participate in information aggregation and introduce spurious dependencies.

C. Invariant Learning and Intervention-Based Robustness

Invariant learning aims to capture predictive mechanisms that remain stable across environments [18], [19], [20], while causal intervention theory studies distributional changes induced by external manipulations [21], [22]. However, existing invariant learning methods often require multiple environments or predefined environment partitions, which are difficult to obtain in dynamic stock markets. Inspired by causal invariance, DualPerturb constructs targeted temporal and relational perturbations from potentially confounding components and low-relevance relations, respectively, encouraging stable predictions without requiring explicit environment labels or claiming causal identification.

Algorithm 1 Training pipeline of DualPerturb

Input: Historical stock indicators $\mathbf{X}$ and labels $\mathbf{Y}$.
Output: Stock movement predictions $\widehat{\mathbf{Y}}$.
for each training batch **do**
  # 1. Temporal Encoding.
  Partition each stock indicator into temporal patches.
  Encode and aggregate the patches to obtain $\mathbf{H}$.
  # 2. Temporal Decomposition.
  Generate complementary masks $\mathbf{M}_P^t$ and $\mathbf{M}_F^t$.
  Decompose $\mathbf{H}$ into the predictive component $\mathbf{H}_P$ and the potentially confounding component $\mathbf{H}_F$.
  # 3. Stock Relationship Learning.
  Compute pairwise relation scores $\mathbf{S}$ from $\mathbf{H}_P$.
  Retain high-relevance relations to construct $\mathbf{A}_P$.
  Construct the perturbation relations $\mathbf{A}_F$ from the excluded low-relevance relations.
  Aggregate the retained relations to obtain the predictive spatio-temporal representation $\mathbf{Z}_P$.
  # 4. Dual Perturbation Learning.
  Reintroduce $\mathbf{H}_F$ into $\mathbf{Z}_P$ to construct the temporal perturbation $\mathbf{Z}_T^{\mathrm{pt}}$.
  Inject $\mathbf{A}_F$ into $\mathbf{Z}_P$ to construct the relational perturbation $\mathbf{Z}_R^{\mathrm{pt}}$.
  # 5. Prediction and Optimization.
  Apply the shared predictor to $\mathbf{Z}_P$, $\mathbf{Z}_T^{\mathrm{pt}}$, and $\mathbf{Z}_R^{\mathrm{pt}}$.
  Compute the prediction loss $\mathcal{L}_P$, temporal perturbation loss $\mathcal{L}_T$, and relational perturbation loss $\mathcal{L}_R$.
  Jointly optimize the model using $\mathcal{L}_P$, $\mathcal{L}_T$, and $\mathcal{L}_R$.
end for
return $\widehat{\mathbf{Y}}$ predicted from $\mathbf{Z}_P$.

II. Theoretical Analysis

Proposition II.1. *Let $\mathbf{Z}_P$ denote the predictive spatio-temporal representation for predicting $\mathbf{Y}$, and let $\mathbf{Z}_F$ denote the potentially confounding information that provides no additional predictive information about $\mathbf{Y}$ given $\mathbf{Z}_P$. Assume that $\mathbb{E}\|\mathbf{Y}\|_2^2 < \infty$ and $\mathbb{E}[\mathbf{Y} \mid \mathbf{Z}_P, \mathbf{Z}_F] = \mathbb{E}[\mathbf{Y} \mid \mathbf{Z}_P]$. Define the optimal predictor as $m(\mathbf{Z}_P) = \mathbb{E}[\mathbf{Y} \mid \mathbf{Z}_P]$. For any square-*

integrable predictor $f(\mathbf{Z}_P, \mathbf{Z}_F)$*, we have*

$$\begin{aligned}&\mathbb{E}\left[\|\mathbf{Y} - f(\mathbf{Z}_P, \mathbf{Z}_F)\|_2^2\right] \\ &= \mathbb{E}\left[\|\mathbf{Y} - m(\mathbf{Z}_P)\|_2^2\right] + \mathbb{E}\left[\|f(\mathbf{Z}_P, \mathbf{Z}_F) - m(\mathbf{Z}_P)\|_2^2\right].\end{aligned} \tag{1}$$

Consequently,

$$\mathbb{E}\left[\|\mathbf{Y} - f(\mathbf{Z}_P, \mathbf{Z}_F)\|_2^2\right] \geq \mathbb{E}\left[\|\mathbf{Y} - \mathbb{E}\left[\mathbf{Y} \mid \mathbf{Z}_P\right]\|_2^2\right]. \tag{2}$$

Proof. Let $m(\mathbf{Z}_P) = \mathbb{E}[\mathbf{Y} \mid \mathbf{Z}_P]$. Expanding the squared prediction error gives

$$\begin{aligned}&\mathbb{E}\left[\|\mathbf{Y} - f(\mathbf{Z}_P, \mathbf{Z}_F)\|_2^2\right] = \mathbb{E}\left[\|\mathbf{Y} - m(\mathbf{Z}_P)\|_2^2\right] \\ &+ \mathbb{E}\left[\|f(\mathbf{Z}_P, \mathbf{Z}_F) - m(\mathbf{Z}_P)\|_2^2\right] \\ &+ 2\mathbb{E}\left[\langle \mathbf{Y} - m(\mathbf{Z}_P), m(\mathbf{Z}_P) - f(\mathbf{Z}_P, \mathbf{Z}_F)\rangle\right].\end{aligned} \tag{3}$$

Since $m(\mathbf{Z}_P) - f(\mathbf{Z}_P, \mathbf{Z}_F)$ is $\sigma(\mathbf{Z}_P, \mathbf{Z}_F)$-measurable, we have

$$\begin{aligned}&\mathbb{E}\left[\langle \mathbf{Y} - m(\mathbf{Z}_P), m(\mathbf{Z}_P) - f(\mathbf{Z}_P, \mathbf{Z}_F)\rangle\right] \\ &= \mathbb{E}\left[\langle \mathbb{E}\left[\mathbf{Y} - m(\mathbf{Z}_P) \mid \mathbf{Z}_P, \mathbf{Z}_F\right], m(\mathbf{Z}_P) - f(\mathbf{Z}_P, \mathbf{Z}_F)\rangle\right].\end{aligned} \tag{4}$$

Based on the assumption $\mathbb{E}[\mathbf{Y} \mid \mathbf{Z}_P, \mathbf{Z}_F] = m(\mathbf{Z}_P)$, we obtain $\mathbb{E}\left[\mathbf{Y} - m(\mathbf{Z}_P) \mid \mathbf{Z}_P, \mathbf{Z}_F\right] = \mathbf{0}$. Therefore, the cross term vanishes and

$$\begin{aligned}&\mathbb{E}\left[\|\mathbf{Y} - f(\mathbf{Z}_P, \mathbf{Z}_F)\|_2^2\right] \\ &= \mathbb{E}\left[\|\mathbf{Y} - m(\mathbf{Z}_P)\|_2^2\right] + \mathbb{E}\left[\|f(\mathbf{Z}_P, \mathbf{Z}_F) - m(\mathbf{Z}_P)\|_2^2\right].\end{aligned} \tag{5}$$

Since the second term is non-negative, the result follows. □

Proposition II.1 shows that relying on non-predictive confounding information cannot reduce the optimal prediction risk and may instead incur excess error. This motivates us to identify the predictive components of stock spatio-temporal representations for reliable prediction. The proof is provided in the supplementary material due to space limitations.

III. Experiment

A. Dataset

To comprehensively evaluate the effectiveness of our model across different stock markets, we conduct experiments on three representative datasets from the Chinese and U.S. markets: CSI 300, CSI 500, and S&P 500. After preprocessing to ensure data completeness, the resulting datasets are summarized as follows:

- CSI 300 & CSI 500: After excluding stocks with substantial missing trading records, primarily caused by recent listings or trading suspensions, 176 and 284 constituent stocks are included, respectively.
- S&P 500: Following the same preprocessing procedure, 478 of the 500 constituent stocks are retained after excluding stocks with incomplete trading records.

All stock data are collected from TuShare[1]. CSI 300 and CSI 500 cover 1,699 trading days from January 2019 to December 2025, while S&P 500 covers 1,195 trading days from January 2019 to September 2023. After removing stocks with incomplete trading records, primarily due to trading suspensions, 176, 284, and 478 stocks are retained, respectively. Following prior studies, stock movement prediction is formulated as a binary classification task of upward and downward movements. All datasets are chronologically divided into training, validation, and test sets at a ratio of 70%, 15%, and 15%, respectively. Table I presents the detailed statistics.

[1]https://www.tushare.pro/

	CSI 300	**CSI 500**	**S&P 500**
# stocks	176	284	478
Period	2019.1-2025.12	2019.1-2025.12	2019.1-2023.9
# Trading Days	1699	1699	1195
Train:Val:Test	1169:255:255	1169:255:255	816:179:180

TABLE I: Datasets Statistics.

B. Metrics

Following prior studies [8], [9], we adopt accuracy (ACC) and Matthews correlation coefficient (MCC) to evaluate prediction performance. ACC measures the proportion of correctly classified samples among all test samples, while MCC provides a more balanced assessment by considering all four entries of the confusion matrix and is particularly suitable when the class distribution is imbalanced. The two metrics are defined as follows:

$$\text{ACC} = \frac{TP + TN}{TP + TN + FP + FN}, \tag{6}$$

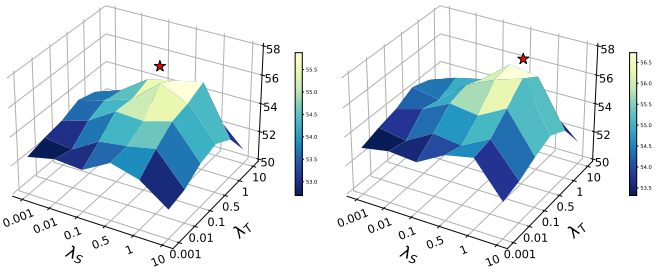
$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}. \tag{7}$$

where TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively.

C. Baselines

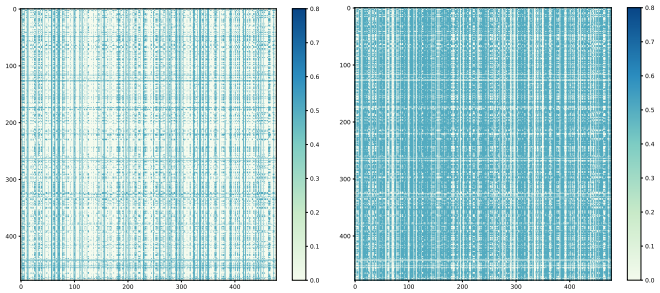
We compare DualPerturb with eight representative stock prediction baselines, which are briefly introduced as follows:

- GAT [23]: GAT employs self-attention to assign importance weights to related stocks during graph information aggregation. It enables different stocks to contribute unequally to stock representation learning without relying on fixed aggregation weights.
- TGC [24]: TGC jointly models the temporal evolution of individual stocks and their relational dependencies through temporal graph convolution.
- ADGAT [14]: ADGAT introduces attribute-driven graph attention to capture momentum spillovers among stocks and employs unmasked attention to infer dynamic inter-stock relations from market signals.
- MGAR [25]: MGAR constructs multiple stock relation graphs, including a dynamic price-similarity graph and a predefined industry relation graph, to capture complementary inter-stock dependencies.



(a) CSI 300 dataset. (b) S&P 500 dataset.

Fig. 1: Hyperparameter sensitivity to λ_T and λ_R.



(a) Predictive relations $\mathbf{A}_P$. (b) Perturbation relations $\mathbf{A}_F$.

Fig. 2: Visualization of the inter-stock relations on CSI 300.

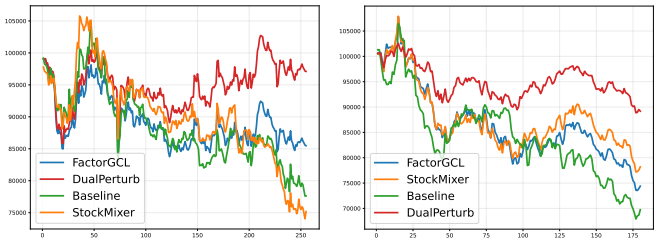
- VGNN [15]: VGNN models vague stock graphs by jointly considering explicit and implicit relations, and employs attribute-sensitive message passing to capture momentum spillovers among stocks.
- MDGNN [26]: MDGNN constructs a multi-relational dynamic graph to capture heterogeneous inter-stock dependencies and their temporal evolution. It further employs a Transformer-based architecture to encode the evolving multiplex relations for stock investment prediction.
- StockMixer [16]: StockMixer is an MLP-based architecture that sequentially models indicator, temporal, and cross-stock correlations through dedicated mixing operations. It performs multi-scale temporal mixing and explicitly models stock-to-market and market-to-stock interactions for stock forecasting.
- FactorGCL [27]: FactorGCL employs a hypergraph-based factor model to capture high-order nonlinear relationships between stocks. It further introduces a cascading residual architecture and temporal residual contrastive learning to extract hidden factors for stock prediction.

D. Hyperparameter Sensitivity

We investigate the sensitivity of DualPerturb to the temporal and relational perturbation weights λ_T and λ_R, respectively. As shown in Fig. 1, the model generally achieves better performance when both coefficients are between 0.1 and 1, while excessively small or large values lead to performance degradation. Specifically, the best performance on CSI 300 is obtained at $\lambda_T = 0.5$ and $\lambda_R = 0.5$, whereas S&P 500 achieves its optimum at $\lambda_T = 0.5$ and $\lambda_R = 1$. The relatively larger λ_R on S&P 500 may be attributed to its larger number of stocks, which involves more complex cross-stock interactions and may therefore benefit from stronger relational perturbations. Moreover, the relatively stable performance within this range demonstrates that DualPerturb is not highly sensitive to these hyperparameters.

E. Matrix Visualization Analysis

Fig. 2 visualizes the learned predictive and perturbation relation matrices on the CSI 300 dataset. We observe that $\mathbf{A}_P$ exhibits a sparse rather than fully connected structure, indicating that only high-relevance inter-stock relations are retained for prediction. In contrast, $\mathbf{A}_F$ presents a largely complementary pattern, in which relations excluded from $\mathbf{A}_P$ receive larger perturbation weights. This construction generates targeted relational perturbations from low-relevance relations, exposing the predictor to potential relational interference during training. These observations show that DualPerturb differentiates high-relevance relations from low-relevance ones instead of treating all inter-stock dependencies uniformly.



(a) CSI 300 dataset. (b) S&P 500 dataset.

Fig. 3: Back-testing performance illustration.

F. Investing Simulation

To further evaluate the effectiveness of the proposed model in practical investment scenarios, we conduct an investment simulation on the test set, as shown in Fig. 3. We assume that the market is sufficiently liquid and that all trades are executed at closing prices. For simplicity, portfolio returns are calculated based on equal-weight allocations, abstracting away from minimum trading-unit constraints. At the close of each trading day, stocks predicted to move upward are purchased with equal capital allocations and sold at the close of the following trading day. A transaction cost of 0.1% is applied to each complete buy-and-sell transaction. Moreover, an equal-weight portfolio comprising all eligible stocks is employed as the market baseline. The results demonstrate that DualPerturb outperforms the strongest competing methods, FactorGCL and StockMixer, as well as the equal-weight market baseline on both the CSI 300 and S&P 500 datasets, while maintaining more stable portfolio value curves. These findings suggest that, by reducing reliance on weak or spurious inter-stock dependencies, DualPerturb generates more reliable stock-prediction signals and delivers superior practical investment performance.

IV. Discussion and Future Work

The results suggest that temporal and relational perturbations help DualPerturb capture stable predictive information and reduce its sensitivity to spurious dependencies. Nevertheless, the method is motivated by causal invariance rather than causal identification. Future work will explore explicit invariance constraints and robustness under market regime shifts.

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